

Forensic investigation

In case a computer system was compromised, following questions can get the investigation started:

- Was the system compromised?
- How did the breach occur?
- What systems were compromised?
- What did the hackers take?
- What did they change?
- What kind of attack was it?
- Whether it was a new attack?
- What is the trend or the cause?
- How do we remedy the incident?
- How much is the loss or damage to the system?
- What were the tools and methods used to attack the system?



Fig. 5: Branches of digital forensics

and solve issues like identity of the owner, purpose of data stolen and so on.

Reporting. Finally, evidence is presented in the form of a written report when forensic investigation is completed. Generating evidence in the form of a report is crucial. Every task in the forensic analysis has to be written or documented carefully. The investigator should be able to explain complex technological concepts in simple terms.

Fig. 4 presents a typical computer forensic investigation strategy.

Legal considerations

In a research environment, the primary goal is to learn and not to prosecute, so legal considerations and data handling, for example, is not as important. But in corporate or law enforcement investigation, restrictions against network based system monitoring or reading of personal

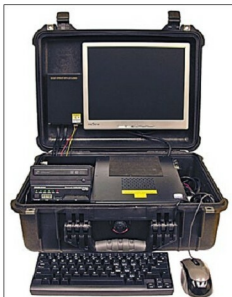


Fig. 6: Forensic Air-Lite V MK III, a forensic computer system (Image courtesy: forensiccomputers.com)

communications often exist.

During a criminal investigation, there is restriction on the amount of information that can be seized as per national laws. For example, in the UK, seizure of evidence by law enforcement is governed by Police and Criminal Evidence Act, 1984 (PACE Act). International Organisation on Computer Evidence (IOCE) is one agency that works to establish compatible international standards for the seizure of evidence.

Key concepts during investigation

Digital forensic uses a set of techniques and procedures that have to be developed during an investigation process. Following are some key concepts to consider in a forensic technique:

Never analyse the original. Do not make use of the original system for investigation. Always make an image or copy of the system logs and analyse that. If you find that somewhere an image is not the same as the original, try to restore it using the original. This point does not apply if you are working on a live system; skip the point instead. Install as little as possible on the system if you are going for live forensics.

Do not trust the suspect system.



Fig. 7: FS-NyarL, a network forensic tool (Image courtesy: fulgursecurity.com)

Never trust any suspicious process or application on the system. If you need to execute programs on that system, try to use a CD of trusted binaries.

Generate hash values of all data. Try to take hash values (using MD5 or SHA algorithm) of the original data; this will help in verification of the integrity of data in future investigations.

Document everything. Try to list everything like modified entries and installed logs, so that you can filter out such activities from system logs.

Platform based analysis

Many tools can analyse both Linux and Windows systems, so you are not restricted to the platform of the compromised systems. But there are tools that are restricted to platform types. In such cases, categorise forensic tools based on platform types. There are commercial tools as well as free open source tools available in the market. Many open source and free tools offer the same basic functionalities as commercial tools.

1. Linux based forensics (specific for Linux systems only)
2. Windows based forensics (specific for Windows systems only)
3. Hybrid forensics (includes both Linux and Windows systems)